

Module M8L: Morning Star D20 -- Operational Certification

Battle Plan v1.6 | David Fox | May 21, 2026

First Transit | Full Hub Open | Commercial Ops | Return Route | Health Check | Round-Trip Certified

STATUS: MORNINGSTAR_OPERATIONAL_CERTIFIED -- All 9 checks PASS

OPS-1: H01->Proxima Dock, $t=7.71$ ns, 4.24 ly, tidal=0.092 g < 0.1 g. TRANSIT_SUCCESS.

OPS-2: HUB_FULL_OPEN. 12 faces, 30 routes, 1260 kW hold, 2740 kW margin.

OPS-3: 12 destinations. Proxima 7.71 ns to Kepler-442 92.52 ns. All $t=n*t_{hop}$ PASS.

OPS-4: COMMERCIAL_SUCCESS. 47 tx/hr, 312 pax/hr, 89 t/hr, 604.3 ly/hr.

OPS-5: DOCK_A<->H01 bidirectional. 31/120 destinations registered. RETURN_SUCCESS.

OPS-6: HEALTH_PASS. 120/120 cells, 1680/1680 PLL, TDC=3.001 ps, cryo=4.003 K.

OPS-7: RETURN_SUCCESS + HEALTH_PASS. Full loop confirmed. abort_flag=0.

STDOUT SHA-256

80ff8a251c6ea7b6a57fd81fe71a76dd62a3f862c80381d571e2f30d3c4222ad

SOURCE SHA-256 (certificates/m8l_morningstar_ops.py)

8ad0a5b91c8182d9d8264d8ff776834fdce28469abd778078a4e31309353f525

1. Overview

Module M8L certifies the full operational commissioning of MORNING_STAR_D20, the first wormhole starport constructed using the certified FTL Morningstar stack (M8K). This module records seven operational milestones (OPS-1 through OPS-7) spanning first light, commercial traffic, return-route registration, full health check, and round-trip loop closure.

All numerical inputs are SHA-bound certified values from the upstream chain. No free parameters are introduced. The hub geometry (dodecahedron, 12 faces, 30 routes, 120-cell fabric) follows directly from the D20 design specification.

2. Causal DAG

Module	Certified claim	Input to M8L
M8J	$\Delta t=7.647$ ns, tidal<0.1 g (OQs closed)	t_{hop} base, tidal limit
M8K	$t_{hop}=7.71$ ns, RTT=18.635 ns, $B_M=21.768$ MHz	operational hop slot
M8I	14-mode Nb3Sn H4 resonator, 120-cell fabric	pll_per_cell=14, cells=120

3. Hub Geometry: MORNING_STAR_D20

The hub is a physical realisation of the dodecahedron (dual of icosahedron, hence D20). Twelve pentagonal faces correspond to the twelve hallway faces H01-H12. Thirty edges give thirty one-face-to-one-face routes at full open. The Euler relation $V - E + F = 20 - 30 + 12 = 2$ confirms closed-manifold topology (PASS).

The switching fabric is the HYPER120_001 core: 120 cells, each hosting 14 PLL chains from the certified 14-mode Nb3Sn H4 resonator (M8I/M8J). Total PLL population: $14 \times 120 = 1680$.

4. Operational Milestones

OPS-1: First Public Transit

Command: morning_star_transit.py --src_face H01 --dest "Proxima Dock" --personnel 1 --event_name FIRST_PUBLIC_TRANSIT

Parameter	Value
Route	H01 -> Proxima Dock
Distance	4.24 ly
Hops	1
t_transit	7.71 ns (= 1 x t_hop, PASS)
Tidal at throat	0.092 g (< 0.1 g OQ-1 limit, PASS)
sha_chain	[62492d66, 2c3ac1d2]
Result	TRANSIT_SUCCESS

OPS-2: Full Hub Open (30 Routes)

Command: morning_star_traffic.py --routes 30 | hyper120_arm.py --cascade_s 14 --parallel

```

routes_armed = 30 / 30
faces_active = 12 / 12
power_hold   = 30 x 42 kW = 1260 kW
generation   = 2 MW solar + 2 MW RTG = 4000 kW
margin       = 4000 - 1260 = 2740 kW (PASS)
Status       : HUB_FULL_OPEN

```

OPS-3: 12-Destination Manifest

All 12 destination transit times satisfy $t = n \times t_{\text{hop}}$ within 0.03 ns tolerance. Wolf 1061 and Gliese 667C display 23.15 ns (CONOPS rounding) vs computed $3 \times 7.71 = 23.13$ ns; delta = 0.02 ns < 0.1% of t_{hop} -- rounding artefact only.

Destination	Dist (ly)	Hops	t (ns)	Check
Proxima Dock	4.24	1	7.71	PASS
Barnard's Star	5.96	2	15.42	PASS
Wolf 1061	14.20	3	23.15	PASS*
Tau Ceti	11.90	2	15.42	PASS
Epsilon Eri	10.48	2	15.42	PASS
Ross 128	11.01	1	7.71	PASS
Luyten's Star	12.35	2	15.42	PASS
Teegarden's	12.52	2	15.42	PASS
Trappist-1	40.66	4	30.84	PASS
LHS 1140	40.67	4	30.84	PASS
Gliese 667C	23.62	3	23.15	PASS*
Kepler-442	1193.0	12	92.52	PASS

* PASS* = 23.15 ns vs 23.13 ns computed; delta=0.02 ns CONOPS rounding (< 0.1% of t_{hop}).

OPS-4: Commercial Operations (1-Hour Snapshot)

```

Transits / hr   : 47
Personnel / hr  : 312
Cargo / hr      : 89 t
Light-years / hr : 604.3

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Energy / hr : 9.4 MWh
Energy / person : 16.7 kWh
Abort flag : 0 Uptime : 100.0%
Result : COMMERCIAL_SUCCESS

OPS-5: Return Route Registration

```
add_route.py --name "Sol - Lamma Dock" --face DOCK_A
              --cell 000 --dist_ly 0.0000045 --type bidirectional
ROUTE_ADDED: Sol - Lamma Dock | DOCK_A | 7.71 ns | 0.0000045 ly
REGISTRY : 31 / 120 destinations online
Return test: DOCK_A -> H01 | t=7.71 ns | RETURN_SUCCESS
```

OPS-6: Full Health Check (120 Cells)

Parameter	Measured	Limit	Check
Cells online	120 / 120	= 120	PASS
PLL locked	1680 / 1680	= 1680	PASS
TDC resolution	3.001 ps	< 4 ps	PASS
Cryo temperature	4.003 K	< 4.5 K	PASS
RF Q-factor	9.8e9	> 1e9	PASS
Failures	[]	= 0	PASS

OPS-7: Round-Trip Certification

RETURN_SUCCESS + HEALTH_PASS = full loop confirmed.
All 120 cells green. abort_flag = 0. Uptime = 100%.
Morning Star is round-trip certified.

5. Pass/Fail Summary

Check	Result
Euler V-E+F=2 (dodecahedron geometry)	PASS
t_hop consistency: t_hop - Delta_tau / Delta_tau < 1%	PASS
First transit time = t_hop (H01->Proxima, 7.71 ns)	PASS
Tidal at throat < 0.1 g (OQ-1, M8J certified)	PASS
Power margin > 0 (4000 kW gen - 1260 kW hold)	PASS
12-dest manifest: all t = n * t_hop (tol 0.03 ns)	PASS
Commercial throughput: ly/hr > 0	PASS
Return route: DOCK_A->H01 t = t_hop	PASS
Health: cells, PLL, TDC, cryo, RF all within limits	PASS
ALL 9 CHECKS	PASS

6. Certified Values

Module: M8L -- Morning Star D20 Operational Certification
Causal parents: M8J (Delta_tau=7.647 ns, tidal<0.1 g)
M8K (t_hop=7.71 ns, RTT=18.635 ns, B_M=21.768 MHz)
Source SHA-256: 8ad0a5b91c8182d9d8264d8ff776834fdce28469abd778078a4e31309353f525
Stdout SHA-256: 80ff8a251c6ea7b6a57fd81fe71a76dd62a3f862c80381d571e2f30d3c4222ad

```
=== HUB GEOMETRY ===
MORNING_STAR_D20 dodecahedron 12 faces 30 routes 20 vertices
Euler V-E+F = 20-30+12 = 2 (PASS)
Core: HYPER120_001 120 cells 14 PLL/cell 1680 PLL total

=== FIRST PUBLIC TRANSIT (OPS-1) ===
H01 -> Proxima Dock 4.24 ly t=7.71 ns tidal=0.092 g TRANSIT_SUCCESS

=== FULL HUB (OPS-2) ===
30 routes 12 faces 1260 kW hold 2740 kW margin HUB_FULL_OPEN

=== 12-DESTINATION MANIFEST (OPS-3) ===
Proxima Dock 7.71 ns Barnard's 15.42 ns Wolf 1061 23.15 ns
Tau Ceti 15.42 ns Eps Eri 15.42 ns Ross 128 7.71 ns
Luyten's 15.42 ns Teegarden 15.42 ns Trappist-1 30.84 ns
LHS 1140 30.84 ns Gl 667C 23.15 ns Kepler-442 92.52 ns

=== COMMERCIAL OPS (OPS-4) ===
47 tx/hr 312 pax/hr 89 t/hr 604.3 ly/hr 9.4 MWh/hr COMMERCIAL_SUCCESS

=== RETURN ROUTE (OPS-5) ===
Sol - Lamma Dock DOCK_A 0.0000045 ly bidirectional 31/120 RETURN_SUCCESS

=== HEALTH CHECK (OPS-6) ===
120/120 cells 1680/1680 PLL TDC=3.001 ps cryo=4.003 K Q=9.8e9 HEALTH_PASS

=== ROUND-TRIP CERTIFICATION (OPS-7) ===
RETURN_SUCCESS + HEALTH_PASS abort_flag=0 uptime=100%
Morning Star is round-trip certified.

ALL 9 CHECKS PASS
STATUS: MORNINGSTAR_OPERATIONAL_CERTIFIED
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THEOREM M8L (axiom_debt: [], status: MORNINGSTAR_OPERATIONAL_CERTIFIED):

Hub MORNING_STAR_D20 is a fully commissioned wormhole starport. Geometry: dodecahedron, 12 faces, 30 routes, 120-cell HYPER120_001 fabric, 1680 PLL chains (Euler V-E+F=2 PASS). OPS-1: H01->Proxima Dock, t=7.71 ns, 4.24 ly, tidal=0.092 g < 0.1 g (OQ-1, PASS). OPS-2: HUB_FULL_OPEN, 30 routes, 1260 kW hold, 2740 kW margin (PASS). OPS-3: 12 destinations Proxima (7.71 ns) through Kepler-442 (92.52 ns), all t=n*t_hop within 0.03 ns (PASS). OPS-4: COMMERCIAL_SUCCESS, 47 tx/hr, 312 pax/hr, 89 t/hr, 604.3 ly/hr (PASS). OPS-5: DOCK_A<->H01 bidirectional registered, 31/120 destinations, RETURN_SUCCESS (PASS). OPS-6: HEALTH_PASS, 120/120 cells, 1680/1680 PLL, TDC=3.001 ps, cryo=4.003 K, Q=9.8e9, failures=[] (PASS). OPS-7: full loop confirmed, abort_flag=0, uptime=100%, Morning Star is round-trip certified (PASS). All 9 checks PASS. Causal parents: M8J (Delta_tau, tidal bound), M8K (t_hop, RTT, B_M). No free parameters introduced.